

AP02 clearance and free-fraction assessment, recovery-controlled dialysis

Final Report

Study MV-2018-052 (LIMS M18/BIO/01223)

Client: Avelnelwick Pharma

Study Director: Dr M. Larkin

Report date: April 16, 2018

1 Summary

40 compounds from the Avelnelwick aminopyrimidine regioisomer series 02 were tested for Avelnelwick Pharma in the Microsomal CL_{int}, fu,p by equilibrium dialysis. Testing started on March 12, 2018 and results were reported on April 16, 2018. All runs met matrix acceptance criteria.

Table 1 Study summary

Metric	Value
Project	MV-2018-052
Client	Avelnelwick Pharma
Compounds tested	40
Series / chemotype	Avelnelwick aminopyrimidine regioisomer series 02 (Aminopyrimidine regioisomer)
Assays	Liver microsomal stability, Plasma protein binding (equilibrium dialysis)
Turnaround (calendar days)	35
Price	USD 23,500

2 Objective

To rank the test article set for Microsomal CL_{int}, fu,p by equilibrium dialysis liability so that Avelnelwick Pharma could choose analogues for progression.

3 Test items

40 compounds were received as 10 mM DMSO solutions in one accession lot on March 12, 2018 and logged against M18/BIO/01223. Identity and concentration were taken as supplied by the client.

4 Methods

Microsomal CLint

Test articles (1 μ M) were incubated with pooled human and rat liver microsomes (0.5 mg/mL protein) and NADPH at 37 °C. Samples at 0, 5, 15, 30 and 45 minutes were quenched in acetonitrile with internal standard and analysed by LC-MS/MS. Intrinsic clearance was calculated from the first-order depletion rate; half-lives above 60 minutes are reported as >60.

Matrix acceptance: verapamil (high clearance) and diltiazem (moderate clearance) within historical range in each run.

fu,p by equilibrium dialysis

Binding to human plasma was measured by rapid equilibrium dialysis (5 μ M test article, 4 h at 37 °C, PBS pH 7.4). Plasma and buffer compartments were matrix-matched before LC-MS/MS. Fraction unbound (fu) is reported with mass-balance recovery; recoveries below 70% are footnoted.

Warfarin was included as the binding control in every plate.

5 Results

Microsomal stability

Table 2 Microsomal intrinsic clearance and half-life

Compound	Human CLint (μ L/min/mg)	Human t $\frac{1}{2}$ (min)	Rat CLint (μ L/min/mg)	Rat t $\frac{1}{2}$ (min)
AP02-01404	19	>60	47	29
AP02-01406	23	59	40	35
AP02-01412	21	>60	36	39
AP02-01416	18	>60	25	55
AP02-01418	58	24	113	12
AP02-01422	20	>60	36	39
AP02-01434	27	51	46	30
AP02-01440	18	>60	18	>60
AP02-01448	15	>60	32	43
AP02-01452	16	>60	38	37
AP02-01454	27	52	44	32
AP02-01460	21	>60	26	53
AP02-01464	25	55	48	29
AP02-01470	11	>60	15	>60
AP02-01476	11	>60	16	>60
AP02-01478	98	14	186	7.5
AP02-01488	23	>60	34	40
AP02-01494	18	>60	41	34
AP02-01496	11	>60	17	>60
AP02-01500	22	>60	37	38
AP02-01502	14	>60	46	30
AP02-01506	13	>60	11	>60
AP02-01508	4.1	>60	7.0	>60
AP02-01512	9.7	>60	13	>60
AP02-01518	62	22	91	15
AP02-01520	35	39	64	22
AP02-01526	35	39	53	26
AP02-01538	20	>60	28	49

AP02-01542	20	>60	36	39
AP02-01548	24	58	29	48
AP02-01550	21	>60	43	32
AP02-01556	81	17	113	12
AP02-01560	31	45	33	42
AP02-01566	12	>60	13	>60
AP02-01574	8.6	>60	20	>60
AP02-01578	34	41	65	21
AP02-01580	40	34	56	25
AP02-01584	20	>60	23	>60
AP02-01586	18	>60	31	45
AP02-01590	25	56	32	44

Controls: verapamil (high), diltiazem (medium) within historical range

Plasma protein binding

Table 3 Human plasma protein binding

Compound	% bound	fu	Recovery (%)
AP02-01404	90.8	9.18	81
AP02-01406	90.3	9.73	99
AP02-01412	99.9	0.10	86
AP02-01416	79.6	20.43	82
AP02-01418	94.3	5.71	82
AP02-01422	81.6	18.44	88
AP02-01434	96.9	3.10	91
AP02-01440	93.0	7.04	102
AP02-01448	89.7	10.32	84
AP02-01452	97.2	2.80	82
AP02-01454	96.0	3.97	89
AP02-01460	80.1	19.85	88
AP02-01464	89.6	10.40	103
AP02-01470	89.2	10.84	94
AP02-01476	94.8	5.19	81
AP02-01478	99.9	0.10	98
AP02-01488	94.4	5.63	93
AP02-01494	99.5	0.50	93
AP02-01496	83.6	16.43	93
AP02-01500	93.0	6.96	83
AP02-01502	93.7	6.26	88
AP02-01506	87.5	12.46	96
AP02-01508	93.9	6.10	97
AP02-01512	88.8	11.22	99
AP02-01518	99.9	0.10	98
AP02-01520	89.8	10.24	90
AP02-01526	94.4	5.65	104
AP02-01538	92.3	7.66	102
AP02-01542	91.2	8.79	88
AP02-01548	99.0	1.04	83
AP02-01550	92.5	7.55	92
AP02-01556	98.7	1.35	100
AP02-01560	88.4	11.61	82
AP02-01566	89.8	10.18	101
AP02-01574	92.7	7.33	91
AP02-01578	99.7	0.33	90
AP02-01580	99.9	0.10	99
AP02-01584	87.6	12.42	96

AP02-01586	95.0	4.99	89
AP02-01590	99.8	0.22	88

Warfarin control 98-99% bound

6 Discussion

Inhibition values are in vitro, at a fixed concentration, and do not by themselves predict clinical interaction risk; free exposure estimates are needed for that assessment.

7 Conclusion

The data support ranking within the set. Recommended next step: carry the compounds without CYP flags into exposure studies and run CYP3A4 IC50s for flagged analogues before in vivo work.

Name	Role	Signature	Date
Megan Larkin	Study Director	<i>Larkin</i>	April 16, 2018
Helen Marchwood	Bioanalytical Quality Manager	<i>Marchwood</i>	April 15, 2018